Department of Information Technology
Gauhati University

Syllabus for
M.Tech. in Information Technology (M.Tech. IT)

Degree: Master of Technology in Information Technology): M.Tech.(IT)

Duration: Two Years (Four Semesters)

Eligibility:

1. Engineering/Technology Graduates (4 year) in Computer Science & Engineering or Information Technology or Electronics & Telecommunication Engineering or equivalent.

OR,

2. MCA or equivalent.

OR,

3. M. Sc. in Information Technology or Computer Science or Mathematics & Computing or equivalent.

Distribution of Courses for M. Tech. IT:
(Each course may include laboratory work if prescribed)
Total marks for each of the courses listed below will be 100 (end semester examination) except for the course ITC30124 and ITC40124 (Project) .

The Evaluation Patten for each course (except Project Works) is as follows:

1. Class Test (3 Nos) : 50% (50 marks, 15+15+20)
2. End Semester Examination : 50% (100 marks, 3 hour exam)

M.Tech.- Information Technology (IT) Course Structure							
Semester 1							
SL	Course Code	Course Title	L	T	P	C	Contact hrs
1	ITC1016	Machine Learning	3	1	2	6	8
2	ITC1024	Design and Analysis of Algorithms	3	1	0	4	4
3	ITC1034	Seminar Course	0	0	0	2	4
4	ITC1xx4 *	Departmental Elective 1				4	4
5	ITC1xx4 *	Departmental Elective 2				4	4
		Value added Course Offered for other Departments	Audit Course				
		Total=				20	24

Semester 2							
SL	Course Code	Course Title	L	T	P	C	Contact hrs
1	ITC2014	Advance Web Technologies	3	0	1	4	5
2	ITC2024	Cyber Security	3	0	1	4	5
3	ITC2034	Product Development Project.	0	0	0	4	4
4	ITC2xx4 *	Departmental Elective 3				4	4
5	ITC2xx4 *	Departmental Elective 4				4	4
		Value added Course Offered for other Departments	Audit Course				
		Total=				20	22

Semester 3							
SL	Course No	Course Title	L	T	P	C	Contact hrs
1	ITC30124	Project 1	0	0	0	24	0

Semester 4							
SL	Course No	Course Title	L	T	P	C	Contact hrs
1	ITC40124	Project 2	0	0	0	24	0

List of Departmental Electives		
First Semester Electives		
SL	Course Code	Course Name
1	ITC1214	Natural Language Processing
2	ITC1224	Data Mining & Data Warehousing
3	ITC1234	Speech Processing
4	ITC1244	Software Testing and Quality Assurance
5	ITC1254	Nature Inspired Computing
6	ITC1264	Decision Support Systems
7	ITC1274	Advanced System Administration and Networking
8	ITC1284	Wireless Communication and Networks
9	ITC1294	Distributed Systems
10	ITC1304	Applied Graph Theory
11	ITC1314	Educational Technologies
12	ITC1324	E-learning Technologies
13	ITC1334	Electronic Commerce Technologies
14	ITC1344	Enterprise Resource Planning
15	ITC1354	Fuzzy Logic and Neural Networks
16	ITC1364	Geographical Information Systems
17	ITC1374	Object Oriented System Design

18	ITC1384	Abstract Algebra
19	ITC1394	Mathematical Foundation
20	ITC1404	Advanced Programming Techniques
21	ITC1414	Advanced Optimization Techniques
Second Semester Electives		
SL	Course Code	Course Name
1	ITC2214	Advanced Database Management systems
2	ITC2224	Advanced Web Technologies
3	ITC2234	Digital Image Processing
4	ITC2244	Cryptography and Computer Security
5	ITC2254	Computer Architecture and Parallel Processing
6	ITC2264	Digital Signal Processing
7	ITC2274	Mobile Computing
8	ITC2284	Pattern Recognition
9	ITC2294	Agent Oriented Computing
10	ITC2304	Artificial Intelligence and Intelligent Systems
11	ITC2314	Internet Protocols
12	ITC2324	Knowledge based systems
13	ITC2334	Linux Kernels
14	ITC2344	Management Information Systems
15	ITC2354	Parallel and Randomized algorithms
16	ITC2364	Real Time Systems
17	ITC2374	Software Project Management
18	ITC2384	Deep Learning
19	ITC2394	Soft Computing
20	ITC2404	Evolutionary Algorithms
21	ITC2414	Advanced Discrete Mathematics
22	ITC2424	Cloud Computing
23	ITC2434	Information Theory & Coding

DETAILED SYLLABUS

ITC 1016 :: Machine Learning

LTPC (3-1-2-6)

Unit-1: Introduction: History of machine learning, Basic concepts

Unit-2:Supervised learning: Supervised learning setup, LMS, Logistic regression, Perceptron, Backpropagation, Learning rules, Activation functions,neural networks, Exponential family, Generative learning algorithms, Gaussian discriminant analysis, Naive Bayes, Support vector machines, Model selection and feature selection, Ensemble methods: Bagging, boosting.

Unit-3:Learning theory: Bias/variance trade-off, Union and Chernoff/Hoeffding bounds, VC dimension, Worst case (online) learning.

Unit-4:Unsupervised learning: Clustering K-means, EM. Mixture of Gaussians, Factor analysis, PCA (Principal components analysis), ICA (Independent components analysis).

Unit-5:Miscellaneous topics: Hypothesis testing, cross-validation, quadratic discriminant Analysis, adaptive hierarchical clustering, gradient boosting

Unit-6: Sensitivity analysis: F1 score, Type 1 error, type 2 error, Bootsraping, Accuracies ,ROC curves, AUC and other quality metrics

Unit-7 : Hybrid Systems: Fuzzy Logic, Genetic Learning, Associative Memories ,SOM etc.

References:

1. Standard books as prescribed by the DRC from time to time on recommendation of the research guides
2. IEEE, ACM and other standard Journals

ITC1024:: Design and Analysis of Algorithms

LTPC (3-1-0-4)

Course Outcome

CO1: Analyse the asymptotic performance of algorithms.

CO2: Design algorithms for different computational problems.

CO3: Demonstrate a familiarity with major algorithms and data structures.

CO4: Apply important algorithmic design paradigms and methods of analysis.

CO5: Synthesize efficient algorithms in common engineering design situations.

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Algorithm Analysis: Concepts in algorithm analysis, time and space complexity, asymptotic notations, Standard notations and common functions, Asymptotic behaviors of polynomials, relative asymptotic growth, ordering by asymptotic growth rates:

Recurrences: substitution method, iteration method, using Master theorem.

Algorithm Design techniques: Divide and Conquer, Dynamic programming, Greedy Algorithm, Back-tracking, Branch and Bound; Illustration of design techniques by application to some specific problems.

Graph Theoretic Algorithms: Basic concepts of graphs, depth-first search and breadth-first search, Minimum spanning tree; Traveling Salesman Problem.

Theory of NP-Completeness: complexity classes – P, NP. co-NP Reducibility and NP-Complete, NP-Hard.

Recommended readings:

1. T. H. Cormen, C. E. Leiserson and R. L. Rivest, Introduction to Algorithms, Tata-Mcgraw Hill Publishers.
2. Horowitz and Sahani, Fundamentals of Computer Algorithms, Galgotia.
3. A. Aho, J. E. Hopcroft and J. D. Ullman, Design and Analysis of Computer Algorithms, Addison-Wesley.
4. S. Baase and Allen Van Gelder, Computer Algorithms-Introduction to Design and Analysis, Pearson Education, LPE.

ITC2014:: Advance Web Technologies

Course Outcome **LTPC (3-0-1-4)**

CO 1: Recall basic protocol and standard for web communication
CO 2: Apply web based programming knowledge for designing interactive web front-end.
CO 3: Design and develop Server-side scripting with database connectivity.
CO 4: Difference between the application and structure of XML, DTD and XSL
CO 5: Explain service oriented web architecture.

Infrastructure for the WWW; Web Protocols and Standards.

Web Client side: functions and working principle of web browsers, conceptual architecture, designing interactive web front-end with HTML, JavaScript and CSS.

Web Server side: web server functionalities, conceptual architecture of web servers, Server-side scripting with database connectivity.

Web Services: Service Oriented Architecture, web services using SOAP, WSDL, UDDI; developing application systems using web services.

Extensible Markup Language (XML): Elements, Attributes and DTD; XSL; Developing XML based web applications.

Advanced Web Technologies: DCOM, CORBA, EJB, AJAX, ISAPI, WAP.

Web Security: Firewalls, Proxy servers, HTPS for secure web communication.

Recommended Readings:

1. Elliotte Rusty Harold; XML 1.1 Bible; Wiley India.
2. Oliver, Dick; SAMS Teach Yourself Html 4 in 24 Hours; Techmedia.
3. Ashbacher, Charles; SAMS Teach Yourself XML in 24 Hours; Techmedia.
4. SAMS Teach Yourself JavaScript in 24 Hours; Techmedia.
5. SAMS Teach Yourself PHP in 24 Hours; Techmedia.
6. Lehnert, Wendy. G.; Web 101 making the 'Net for you; Pearson Education.
7. Sebesta, Robert; World Wide Web Programming.

ITC2024: Cyber Security

LTPC (3-1-0-4)

Overview: Services, Mechanisms and Attacks; The OSI security architecture; Security Models.

Classical Encryption: Symmetric Cipher Model; Substitution Techniques; Block Ciphers and the Data Encryption Standard; Differential and Linear Cryptanalysis; Block Cipher Design Principles; Block Cipher modes of Operation.

Finite Fields: Review of Groups, Rings, Fields and Modular Arithmetic.

Confidentiality using Symmetric Encryption.

Review of Number Theory.

Public Key Cryptography: Principles of Public Key Encryption; the RSA algorithm; Key Management.

Message Authentication and Hash Functions: Authentication Requirements, Functions, Message Authentication Codes, Hash Functions, Hash Algorithms.

Digital Signature and Authentication Protocols: Digital Signatures; Authentication Protocols; Digital Signature Standard.

Authentication Applications: Kerberos.

Email Security: PGP, S/MIME.

IP Security: Overview, IPSec architecture.

Web Security: SSL and Transport Layer Security; HTTPS.

System Security: Intrusion Detection; Malicious Software- Threats and Countermeasures; Firewalls- Design Principles.

Block chain Technology, Application of Block chain

Cyber Awareness, Cyber Ethics, Cyber Law

Recommended Readings:

1. William Stallings; Cryptography and Network Security- Principles and Practice, 3/e; Pearson Education.
2. Atul Kahate; Cryptography and Network Security- 2/e; Tata McGraw Hills.
3. Behrouz A. Forouzan; Cryptography & Network Security; Tata McGraw Hills.
4. Alfred J. Menezes, Paul C. van Oorschot and Scott A. Vanstone; Handbook of Applied Cryptography; CRC Press.

ITC1034:: Seminar Course

LTPC (- - - - - - 4)

Course Outcome:

CO1: Conduct Comprehensive Literature Surveys identify and access relevant literature sources related to their chosen research topics understanding of the chosen area.

CO2: Develop Research Skills to conduct independent research by selecting a topic of interest and engaging in a systematic literature survey.

CO3: Develop Presentation and Communication Skills

CO4: Judge Critical Thinking and Analytical Abilities

CO5: Rewrite the concept and content as a report.

Each student is to choose a research topic of his interest and will perform Literature Survey on the selected topic.

Each student will have to give three seminar presentations and will have to submit a report as per prescribed format at the time of the last one.

The goal is to motivate the students towards research and build their foundation for research works. Emphasis is on a detailed, systematic and comprehensive Literature Survey.

The evaluation pattern will be as follows:

1. Seminar 1 (Introductory) : 20%
2. Seminar 2 (Mid-Term) : 30%
3. Seminar 3 (End-term) : 50%

ITC1214 :: Natural Language Processing

LTPC (3-1-0-4)

Course Outcome

CO1: Identify the natural language processing methods and strategies.
CO2: Extend semantic and syntactic knowledge and its representations of words or sentences of Natural language.
CO3: Compare the strengths and weaknesses of various NLP technologies and frameworks.
CO4: Explain different terms and techniques used by information retrieval system.
CO5: Relate the concept of world Knowledge representation and Reasoning.

Introduction to Natural Language Understanding: The study of languages; Applications; evaluating language understanding systems; the different levels of language analysis; Representations, Understanding, and Organization of Systems.

Syntactic Processing: Linguistic Background; Grammars and Parsing; Features and Augmented Grammars; Grammars for Natural Language; Efficient Parsing; Ambiguity Resolution.

Semantic Interpretation: Semantics and Logical Form; Linking Syntax and Semantics; Ambiguity Resolution.

Context and World Knowledge: Knowledge representation and Reasoning; Using World Knowledge; Discourse Structure; Conversational Agents.

Recommended Readings:

1. James Allen; Natural Language Understanding 2/e; Pearson education.
2. Christopher D. Manning, Hinrich Schiitze; Foundations of Statistical Natural Language Processing; Cambridge, MA: The MIT Press.
3. Daniel Jurafsky, Jame H Martin, James H Martin; Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition; Prentice Hall.

ITC1234 :: Speech Processing LTPC (3-1-0-4)

Course outcome

CO1: Identify a speech signals, its acoustic representation and phonemic behavior.

CO2: Interpreting different types of signals.

CO3: Generate Fourier series, Fourier transform and Filters for a speech signal

CO4: Integrate different speech analysis tool with for speech data.

CO5: Design algorithms for speech synthesis and recognition.

1. Signals: Continuous-time, Discrete-time and digital signals, classification of signals, Energy and power signals, Causal and Non-causal signals, Even and Odd signals.

2. Continuous-time and Discrete-time systems, Static and Dynamic system, Linear and Non-Linear systems, Time invariant and time-variant systems, Stable and unstable systems.

3. Fourier series analysis of Continuous-time Periodic signals, Fourier Transform, Dirac Function, sampling theorem, Anti aliasing filter, signal reconstruction, sampling of BP signals, Z-transform, IIR and FIR filters.

4. Anatomy and physiology of speech Production, spectrographic analysis of speech, categorization of speech sounds, speech modeling, speech complexity, Cepstral and Met cepstral measures, LPC analysis.

5. Intonation and Annotation speech: CART model, Fujisaki's model, Tilt model, KIM-model, ToBI model. Prosoty and Prosoty Boundary: Fujisaki's syntactic phrasing and prosodic phrasing.

Word Boundary: Word segmentation, Daniel-Swingly's model, PARSER model, Bayesian model, Harri's Model, Brent's generative model, Bigram word segmentation, n-gram segmentation, Chinese Restaurant Process(CRP).

6. Speech Coding

Quantization-uniform, Non uniform, Differential quantization, classification of coders, PCM, DPCM, ADPCM, DM parametric or codecs, vocoder, Mixed Excited Linear Predfiction(MELP), Hybrid coders, code Excited LINEAR prediction(CELP), sub-band coding,wide-band speech coders, speech coding standard's: ITU,GSM,MPEG, IETF, TIA, DoD, RCR(japan),ETSI, speech enhancement, speech and speaker recognition.

BOOKS:

1. Speech Signal processing- Thomas F. Quatieri, Pearson Education(Singapore)

2. Speech Production, analysis and coding- P.H. Talukdar, Lamberf Academic Publishing(Germany)

3. Signals and Systems- P. Ramesh Babu, R. Ananda Natarajan, Scitech Publication pvt. Ltd(India)

ITC2234 :: Digital Image Processing

LTPC (3-0-1-4)

Course Outcome:

CO1: Demonstrate the fundamentals of digital image and its mathematical transforms.
CO2: Apply various image enhancement algorithm

CO3: Categorize different image compression method.

CO4: Explain color image processing segmentation and image restoration

CO5: Design and develop image recognition tool.

Introduction: Definition, origin and use of Digital Image Processing.

Digital Image Fundamentals: Elements of Visual Perception, Image Sensing, Acquisition, Sampling, Quantization.

Image enhancement in the Spatial Domain; Image enhancement in the Frequency Domain.

Image restoration; color image processing.

Wavelets and Multiresolution processing.

Image Compression.

Image Segmentation.

Recommended Readings:

1. Rafel C. Gonzalez, Richard E. Woods; Digital Image Processing, 2/e; Pearson Education.
2. Nick Efford; Digital Image Processing: a practical introduction using Java; Addison Wesley/Pearson Education.
3. Douglas A. Lyon; Image Processing in Java; PHI.

ITC2214 :: Advanced Database Management Systems

LTPC (3-0-1-4)

Course Outcome:

CO1: Describe concurrent Query Processing Techniques used in Traditional DBMS

CO2: Explain the organization of data, as per the distributed database

CO3: Analyze an information storage problem and derive an information model

expressed in the form Entity Relation diagram and other optional analysis forms.

CO4: Discuss various system implementation techniques including query processing, optimization, transaction management, concurrency control, database recovery, security, and authentication.

CO5: Summarize the unique challenges and techniques involved in managing image, multimedia, and spatial databases, including content-based indexing and retrieval, as well as specialized indexing techniques such as R-trees, R+ trees, and KD trees.

Review of Relational model, relational algebra, relational calculus and SQL

Review of Semantic modeling: introduction, E-R model, E-R diagrams, design of database with E-R model, Transformation of ER model to relational schema.

Normalization and functional dependencies: design guidelines, functional dependencies; normal forms; dependency-preserving property, lossless join property, algorithms to ensure dependency -preserving property and lossless join property

System implementation techniques: Query processing and optimization; Transaction processing; Concurrency control; Database recovery; Security and authentication;

Object oriented database systems: Concepts of object-oriented databases; Standards, languages and design; Object relational database systems.

Distributed databases: Concepts; Data fragmentation, replication, and allocation techniques; Types of distributed database systems; Query processing in distributed databases; Overview of concurrency control and recovery in distributed databases.

Image, multimedia, and spatial databases: Concepts of Image, multimedia, and spatial databases; Content-based indexing and retrieval, Indexing techniques- R trees, R+ trees, KD trees.

Recommended Readings:

1. R. Elmasri; S. B. Navathe; Fundamentals of Database Systems; Pearson Education.
2. C. J. Date; An Introduction To Database Systems; Pearson Education.

3. D. Stamper, W. Price; Database Design And Management- An Applied Approach; Mcgraw Hill.
4. C. S. R. Prabhu; Object-Oriented Database Systems- Approaches And Architectures; PHI.
5. J. D. Ullman; Principles of Database Systems; Galgotia.

ITC2244 :: Cryptography and Computer Security

LTPC (3-1-0-4)

Course Outcome:
CO1: Explain the importance of computer and network security in day to day life.

To understand the goals or need of cryptography.
To identify and mitigate security vulnerabilities.
To understand Cryptography Theories, Algorithms and Systems.
To know the use and application of different cryptographic algorithms and methods.
To understand necessary Approaches and Techniques to build protection mechanisms in order to secure computer networks.

Part I: Cryptography

Overview: Services, Mechanisms and Attacks; The OSI security architecture; Security Models.

Classical Encryption: Symmetric Cipher Model; Substitution Techniques; Block Ciphers and the Data Encryption Standard; Differential and Linear Cryptanalysis; Block Cipher Design Principles; Block Cipher modes of Operation.

Finite Fields: Review of Groups, Rings, Fields and Modular Arithmetic.

Confidentiality using Symmetric Encryption.

Review of Number Theory.

Public Key Cryptography: Principles of Public Key Encryption; the RSA algorithm; Key Management.

Message Authentication and Hash Functions: Authentication Requirements, Functions, Message Authentication Codes, Hash Functions, Hash Algorithms.

Digital Signature and Authentication Protocols: Digital Signatures; Authentication Protocols; Digital Signature Standard.

Part II: Security Applications and Protocols

Authentication Applications: Kerberos.

Email Security: PGP, S/MIME.

IP Security: Overview, IPSec architecture.

Web Security: SSL and Transport Layer Security; HTTPS.

System Security: Intrusion Detection; Malicious Software- Threats and Countermeasures; Firewalls- Design Principles.

Recommended Readings:

5. William Stallings; Cryptography and Network Security- Principles and Practice, 3/e; Pearson Education.
6. Atul Kahate; Cryptography and Network Security- 2/e; Tata McGraw Hills.
7. Behrouz A. Forouzan; Cryptography & Network Security; Tata McGraw Hills.
8. Alfred J. Menezes, Paul C. van Oorschot and Scott A. Vanstone; Handbook of Applied Cryptography; CRC Press.

ITC1294 :: Distributed Systems

LTPC (3-1-0-4)

Introduction: definition, goals, types.

Architectures: Centralized, decentralized and hybrid systems; Middleware.

Processes: Threads, virtualization, clients, servers, code migration.

Communication: Remote Procedure Call, Message Oriented communication, Stream Oriented communication, Multicast communication.

Naming: Names, Identifiers and Addresses; Flat naming, Structured Naming, Attribute-based naming.

Synchronization: Physical clock synchronization, Logical Clock synchronization; Mutual Exclusion, Election Algorithms.

Consistency and Replication: Introduction; Consistency models and Protocols; Replica Management.

Fault Tolerance: Introductory concepts, failure models. Reliable client-server communication, Reliable group communication; Distributed commit; Recovery.

Distributed Object-based Systems: Architecture, Processes, Communication, Naming, Synchronization.

Distributed File Systems: characteristics of file systems, distributed file system requirements, File service architecture; detailed case study of Sun Network File System (NFS).

Recommended Readings:

1. Tanenbaum & Steen; Distributed Systems Principles and Paradigms, 2/e; Pearson Education.
2. Coulouris, Dollimore & Kindberg; Distributed Systems Concepts and Design, 3/e; Pearson Education.

ITC1334 :: Electronic Commerce Technologies

LTPC (3-0-1-4)

Introduction: Definition, advantages and disadvantages of E-commerce.

Business Models of E-commerce: Models based on relationship of transaction parties; Models based on relationship of transaction types.

Technological infrastructure: The Internet Client-Server applications; software agents; search engine technology; XML; Internet standards and specifications.

E-marketing: traditional marketing versus online marketing; e-advertising, e-branding; Marketing strategies.

E-Security: security on the Internet; Risk management issues.

Cryptography in E-commerce: Public-key encryption, private-key encryption; Hash functions; digital signatures, digital signature certificates; security protocols.

E-Payment systems: Digital payment requirements, risks involved; Digital token-based payment systems; Electronic Cash, Electronic Cheque; Designing e-payment systems; E-payment protocols like iKP.

E-commerce customer relationship management; E-commerce supply chain management.

Mobile commerce: Growth; technologies; WAP.

Legal, ethical and social issues of Electronic Commerce; E-governance.

Recommended Readings:

1. P T Joseph, S. J.; E- commerce: An Indian Perspective 2/e; PHI.
2. Laudon, K. C. & Traver, C. G.; E-Commerce Business, Technology, Society; Addison Wesley.
3. Ince, Darrel; Developing Distributed and E-commerce Applications; Addison Wesley.
4. Ashbacher, Charles; SAMS Teach Yourself XML in 24 Hours; Techmedia.
5. Rayport, Jeffrey & Jaworski, Bernard; ECommerce; Burr ridge, IL: Irwin / McGraw-Hill.
6. Stallings, William; Cryptography and Network Security: principles and Practice; Prentice Hall.
7. The Internet (WWW) is to be used as a source of up-to-date reading materials for topics related to e-commerce.

ITC2314 :: Internet Protocols

LTPC (3-1-0-4)

Introduction: Protocols, Standards and Organizations, Internet Standards, Internet Administration.

Brief review of the OSI model and the TCP/IP Protocol suit; IP versions.

IP Addresses: Classfull and Classless addressing; Delivery, forwarding and routing of IP packets.

Address Resolution Protocols ARP and RARP; The Internet Protocol IP; Internet Control Message Protocol (ICMP); Internet Group Management Protocol (IGMP); User Datagram Protocol (UDP); Transmission Control Protocol (TCP); Stream Control Transmission Protocol (SCTP).

Unicast routing Protocols; Multicast Routing Protocols.

Naming and Configuration: BOOTP, DHCP, NAT, DNS.

Remote Login: TELNET.

File Transfer: FTP, TFTP.

Electronic Mail: SMTP, POP, IMAP.

Network Management: SNMP.

World Wide Web: HTTP.

Multimedia Transmission Protocols: RTP, VOIP.

Recommended Readings:

1. Behrouz A. Forouzan; TCP/IP Protocol Suite, 3/e; Tata McGraw Hill.
2. Eric A. Hall; Internet Core Protocols: The Definitive Guide; O'Reilly.
3. Douglas E. Comer; Internetworking with TCP/IP: Principles, Protocols and Architectures; PHI.

ITC2334 :: Linux Kernel

LTPC (3-0-1-4)

Introduction: overview of Linux Versus Other Unix-Like Kernels, Linux Versions, Basic Operating System Concepts, the Unix Filesystem, Unix Kernels.

Memory Addressing: Memory Addresses, Segmentation, Paging.

Processes: Process Descriptor; Switching, Creating and Destroying Processes.

Interrupts and Exceptions; Timing Measurements.

Memory Management; Process Address Space.

System Calls: System Call Handler and Service Routines.

Signals: The Role of Signals; Sending and Receiving a Signal; Real-Time Signals.

Process Scheduling: Scheduling Policy, Algorithms, System Calls.

Overview of Kernel Synchronization.

The Virtual Filesystem: The Role of the VFS, VFS Data Structures, Filesystem Mounting, Pathname Lookup, Implementations of VFS System Calls.

Managing I/O Devices: I/O Architecture, Associating Files with I/O Devices, Device Drivers.

Disk Caches; Accessing Regular Files; Swapping; The Ext2 Filesystem; Process Communication Program Execution.

Recommended Readings:

1. Daniel P. Bovet, Marco Cesati; Understanding the Linux Kernel; O'Reilly.
2. Robert Love; Linux Kernel Development; Pearson Education, Inc.

3. C. S. Rodriguez, G. Fischer, S. Smolski; The Linux Kernel Primer; Pearson Education.
4. Neil Matthew, Richard Stones; Beginning Linux Programming; Wiley Publishing, Inc.

ITC2364 :: Real Time Systems

LTPC (3-1-0-4)

Basic Real-Time Concepts: Terminology, Real-Time System Design Issues, Example Real-Time Systems.

Hardware Considerations: Basic Architecture, Hardware Interfacing, Central Processing Unit, Memory, Input/Output, Enhancing System Performance.

Real-Time Operating Systems: Real-Time Kernels, Theoretical Foundations of Real-Time Operating Systems, Intertask Communication and Synchronization, Memory Management.

Software Engineering for Real-Time systems: Requirements Specification for Real-Time Systems; Properties of Software, Software System Design.

Performance Analysis And Optimization: Theoretical Preliminaries- Challenges in Analyzing Real-Time Systems; Performance Analysis; I/O Performance; Performance Optimization; Compiler Optimization; Analysis of Memory Requirements; Reducing Memory Utilization.

Engineering Considerations: Faults, Failures, and Bugs; Fault-Tolerance.

Recommended Readings:

1. Phillip A. Laplante; Real-Time Systems Design and Analysis, 3/E; John Wiley & Sons, Inc.
2. Albert M. K. Cheng; Real-Time Systems Scheduling, Analysis, And Verification; John Wiley & Sons, Inc.
3. Rob Williams; Real-Time Systems Development; Butterworth-Heinemann Elsevier)
4. Edited by Mathai Joseph; Real-time Systems Specification, Verification and Analysis; Prentice Hall International, London.
5. Ionescu, Dan.; Real-time Systems: Modeling, Design, and Applications; World Scientific Publishing Co.